Protozoan Parasites - Factoid Sheet

Week 7 Hematology • Lecture 9 • Fisher • memorize-first companion

ScholarRx: Malaria • Protozoa & Ectoparasites

Malaria = mosquito + liver + RBC. Babesia = tick + RBC only. Chagas = kissing-bug feces + muscle/nerve.

Malaria life cycle

Sporozoite injected -> **liver schizont** -> merozoites -> **RBC ring trophozoite** -> trophozoite -> schizont -> rupture/fever -> gametocytes -> mosquito GI sexual cycle -> sporozoites.

Species fingerprints

Species	RBC / fever	Smear / unique clue	Relapse?
Falciparum	Any RBC; often irregular/~48h	Crescent gametocyte; multiple rings; appliqué; schizonts rare	No hypnozoite
Vivax	Retics; 48h	Enlarged RBC + Schüffner dots	**Yes**
Ovale	Retics; 48h	Oval/fimbriated RBC + stippling	**Yes**
Malariae	Old RBC; **72h**	**Band form**, rosette	No hypnozoite
Knowlesi	**24h daily**	May resemble malariae; can be severe	No

Falciparum severity: any-RBC invasion + high parasitemia + PfEMP1/knob-mediated cytoadherence/sequestration -> cerebral malaria, renal failure, pulmonary edema, DIC, severe hemolysis.

Malaria diagnosis + treatment

Problem	Answer
Is malaria present?	**Thick smear** = sensitive concentration
Which species?	**Thin Giemsa smear** = morphology/speciation
Uncomplicated falciparum/unknown in resistant region	**Artemether-lumefantrine** or **atovaquone-proguanil** in Fisher deck
Severe malaria	**IV artesunate** + supportive care
Vivax/ovale relapse prevention	**Primaquine or tafenoquine** for liver hypnozoites; check **G6PD**

Drug mechanisms

- Chloroquine-family: toxic heme handling
- Artemisinins: ROS/multiple targets
- Atovaquone: CoQ analogue / mitochondria
- Proguanil: folate/DHFR pathway

Do NOT miss

- Sexual malaria cycle = mosquito gut
- Falciparum mature schizonts sequester
- Only vivax/ovale = hypnozoite relapse
- Duffy = classic vivax receptor

Babesia + Chagas + Lecture-Only Toxo

Babesiosis

Ixodes tick / transfusion -> RBC only -> hemolytic anemia -> Maltese cross.

Clinical	Fever/chills/myalgia + hemolysis; low Hgb/platelets, high bili/LDH. Severe in older/immunocompromised: ARDS, DIC, renal failure.
Diagnosis	Smear rings/pear forms/**Maltese cross**; PCR; IFA.
Treatment	Mild-moderate: **atovaquone + azithromycin**. Severe: **clindamycin + quinine**.

Malaria vs Babesia

	Malaria	Babesia
Vector	Mosquito	Ixodes tick
Liver stage	**Yes**	**No**
Signature	Species-specific malaria morphology	**Maltese cross**

Chagas disease

Triatomine “kissing bug” defecates near bite -> T. cruzi enters skin/mucosa -> bloodstream trypomastigotes -> intracellular amastigotes in muscle/nerve.

Acute
- Chagoma
- Romaña sign
- Myocarditis
- Meningoencephalitis
- Hepatosplenomegaly

Chronic
- Cardiomyopathy/arrhythmia/CHF
- Sudden death/thromboembolism
- Megaesophagus
- Megacolon

Diagnosis	Acute = microscopy/PCR. Chronic = serology.
Drug	**Benznidazole or nifurtimox**; manage established cardiac/GI sequelae separately.

Toxoplasmosis - slide content, not official Week 7 protozoan LO

- Cat feces oocyst / undercooked-meat tissue cyst
- Most asymptomatic / mono-like
- Chorioretinitis = “headlights in fog”
- AIDS/immunocompromise -> brain abscess

- Congenital: chorioretinitis
- Hydrocephalus
- Intracranial calcifications
- Tx when indicated: pyrimethamine + sulfadiazine

If you see X -> think Y

Crescent gametocyte	**Falciparum**	Band trophozoite + q72h	**Malariae**
Maltese cross + tick	**Babesia**	Romaña + megaesophagus	**Chagas**
Schüffner + relapse	**Vivax/ovale**	Daily malaria	**Knowlesi**

Last 5 minutes: Falciparum = severe/sequesters. Vivax/ovale = hypnozoites. Malariae = band/72h. Knowlesi = daily. Babesia = tick/Maltese cross/no liver. Chagas = bug feces -> heart + megaGI.

Primary source: Fisher 85-slide deck. Malaria Word doc supplements malaria only. Official objectives prioritize malaria, babesiosis, and Chagas; toxoplasmosis is a lecture-only extra.